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Fair Division

Selected Topics in Algorithms and Optimization (Modul V5C4)

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Lecture 4: Nash Welfare and Partial EFX

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Outline of this session. So far we only asked for *fair* allocations. Today we combine fairness with *efficiency*. We define the Nash social welfare and prove that maximising it gives EF1 and Pareto optimal allocations (Caragiannis, Kurokawa, Moulin, Procaccia, Shah, Wang 2016; the same paper introduced EFX). Maximum Nash welfare allocations need not be EFX. However, if we may *donate* some goods, we can get EFX while losing at most a factor 2 in Nash welfare (Caragiannis, Gravin, Huang 2019). Throughout, valuations are **additive**.

Nash Social Welfare

Definition 1 (Nash social welfare)**.** The **Nash social welfare** of an allocation X is the geometric mean of the agents' values,

$$\mathrm{NW}(X) \;=\; \Big(\prod_{i\in N} v_i(X_i)\Big)^{1/n}.$$

An allocation is a **maximum Nash welfare (MNW)** allocation if it maximises NW among all complete allocations. We write $\mathrm{opt} = \max_X \mathrm{NW}(X)$.

The Nash welfare lies between two classical objectives: the *utilitarian* welfare $\sum_i v_i(X_i)$, which ignores fairness, and the *egalitarian* welfare $\min_i v_i(X_i)$, which ignores efficiency. Two properties make it attractive:

- **Scale invariance.** If agent i rescales her valuation to $c\cdot v_i$ with $c > 0$, then every $\mathrm{NW}(X)^n$ is multiplied by c, so the set of MNW allocations does not change. Agents do not need to report their values in a common unit.
- **Balance.** A product is large only if *all* factors are reasonably large; an allocation that gives some agent value 0 has Nash welfare 0.

Assumption. To keep the presentation simple, we assume that some allocation gives every agent a positive value, so $\mathrm{opt} > 0$ and every MNW allocation X satisfies $v_i(X_i) > 0$ for all i. (In general one first maximises the number of agents with positive value and then the product of their values; the results below extend.)

Example 1. Two agents and three goods with values

	a	b	c
agent 1	6	2	2
agent 2	1	1	1

The utilitarian optimum gives a, b, c to agent 1 (welfare 10), leaving agent 2 with nothing. The allocation $X_1 = \{a\}$, $X_2 = \{b, c\}$ has $\mathrm{NW}^2 = 6 \cdot 2 = 12$, and it is the MNW allocation: the other candidates have $\mathrm{NW}^2 \in \{8 \cdot 1,\ 10 \cdot 0,\ 2 \cdot 2,\ 4 \cdot 1\}$ for $X_1 = \{a, b\}$, $\{a, b, c\}$, $\{b\}$, $\{b, c\}$, and the remaining ones are symmetric or worse.

Maximum Nash Welfare Is EF1 and Pareto Optimal

Theorem 1 (Caragiannis, Kurokawa, Moulin, Procaccia, Shah, Wang 2016)**.** *For additive valuations, every MNW allocation is EF1 and Pareto optimal.*

Proof. Let X be an MNW allocation; recall that $v_i(X_i) > 0$ for all i.

Pareto optimality. If Y Pareto dominates X, then $v_i(Y_i) \geq v_i(X_i) > 0$ for all i with at least one strict inequality, so $\mathrm{NW}(Y) > \mathrm{NW}(X)$, a contradiction.

EF1. Suppose agent i envies agent j even after removing any single good, i.e.

$$v_i(X_i) \;<\; v_i(X_j \setminus \{g\}) \qquad \text{for all } g \in X_j. \tag{1}$$

In particular $v_i(X_j) > 0$, so the set $S = \{g \in X_j : v_i(g) > 0\}$ is non-empty. Among the goods in S, choose g^* with the smallest ratio $r_g = v_j(g)/v_i(g)$. We claim

$$\frac{v_j(g^*)}{v_i(g^*)} \;\leq\; \frac{v_j(X_j)}{v_i(X_j)}. \tag{2}$$

To see this, let $\alpha = \min_{g \in S} v_j(g)/v_i(g) = v_j(g^*)/v_i(g^*)$ be the minimum ratio. Then $v_j(g) \geq \alpha\, v_i(g)$ for every $g \in S$, and this also holds for $g \in X_j \setminus S$, since there $v_i(g) = 0$. Summing over all $g \in X_j$ gives $v_j(X_j) \geq \alpha\, v_i(X_j)$, which is (2).

Move g^* from j to i. Only the values of i and j change, and the product of all values is multiplied by

$$\frac{v_i(X_i) + v_i(g^*)}{v_i(X_i)} \cdot \frac{v_j(X_j) - v_j(g^*)}{v_j(X_j)} \;\geq\; \Big(1 + \frac{a}{A}\Big)\Big(1 - \frac{a}{B}\Big),$$

where $a = v_i(g^*) > 0$, $A = v_i(X_i) > 0$, $B = v_i(X_j)$, and we used (2) in the form $v_j(g^*)/v_j(X_j) \leq a/B$. By (1) with $g = g^*$, we have $0 < A < B - a$, hence $\frac{a}{A} > \frac{a}{B-a}$ and

$$\Big(1 + \frac{a}{A}\Big)\Big(1 - \frac{a}{B}\Big) \;>\; \Big(1 + \frac{a}{B-a}\Big)\Big(1 - \frac{a}{B}\Big) \;=\; \frac{B}{B-a} \cdot \frac{B-a}{B} \;=\; 1.$$

So the product strictly increases, contradicting the maximality of X. □

The proof uses additivity in two places: the change of the values when moving one good, and the summation that proves (2). Note also that MNW is not computationally easy: computing an MNW allocation is NP-hard (Exercise 2), even to approximate within some constant factor.

EFX. The same paper introduced EFX, which we saw in Lecture 3: $v_i(X_i) \geq v_i(X_j \setminus \{g\})$ for all i, j and *all* $g \in X_j$. An MNW allocation need not be EFX:

Example 2 (MNW need not be EFX)**.** Two agents and three goods g, p_1, p_2 with $v_1(g) = v_2(g) = 1$, $v_i(p_i) = 1 - \varepsilon$ and $v_i(p_k) = \varepsilon/4$ for $k \neq i$, where $\varepsilon > 0$ is small. The MNW allocations are $X_1 = \{g, p_1\}$, $X_2 = \{p_2\}$ and the symmetric one, with $\mathrm{NW}^2 = (2 - \varepsilon)(1 - \varepsilon)$. Neither is EFX: in the first, $v_2(X_1 \setminus \{p_1\}) = 1 > 1 - \varepsilon = v_2(X_2)$.

It is open whether EFX allocations always exist (even for additive valuations with more than three agents). The way out is to allow *donations*: we give up some goods, but keep a large part of the Nash welfare.

Partial EFX with High Nash Welfare

A **partial allocation** Y allocates only a subset of the goods; the other goods are donated (left unallocated). EFX only compares bundles of agents, so it applies to partial allocations unchanged.

Theorem 2 (Caragiannis, Gravin, Huang 2019)**.** *Let X be an MNW allocation (for additive valuations). Then there is a partial allocation Y with $Y_i \subseteq X_i$ for all i such that Y is EFX and*

$$\mathrm{NW}(Y) \;\geq\; 2^{-(1-1/n)} \cdot \mathrm{opt} \;>\; \tfrac{1}{2}\,\mathrm{opt}.$$

Given X, such a Y can be computed in polynomial time.

In fact we will show the stronger statement $v_i(Y_i) \geq \frac{1}{2} v_i(X_i)$ for every agent i, with equality of Y_i and X_i for at least one agent. The idea: start from X and remove goods from bundles, one at a time, until the remaining bundles can be assigned to the agents in an EFX way.

The algorithm

We maintain bundles $Z_1, \ldots, Z_n$ with $Z_j \subseteq X_j$ (bundle Z_j "belongs" to agent j, but the final assignment may differ). Initially $Z_j = X_j$. A bundle is **touched** if at least one good has been removed from it; let T be the set of touched bundles.

Definition 2 (EFX-feasible, feasibility graph)**.** A bundle Z_j is **EFX-feasible** for agent i if agent i, holding Z_j, would not violate EFX towards any bundle:

$$v_i(Z_j) \;\geq\; v_i(Z_k \setminus \{g\}) \qquad \text{for all bundles } Z_k \text{ and all } g \in Z_k.$$

The **feasibility graph** G is the bipartite graph between agents and bundles with an edge (i, Z_j) if Z_j is EFX-feasible for i and, in case $j \neq i$, additionally $v_i(Z_j) > v_i(Z_i)$.

Edges (i, Z_i) are called **identity edges**. If G has a perfect matching M, then giving every agent i her matched bundle $M(i)$ is an EFX (partial) allocation, since every agent holds an EFX-feasible bundle.

Algorithm 1 (Caragiannis, Gravin, Huang 2019)**.** Start with $Z_j = X_j$ for all j. Repeat:

1. Compute a matching M in G such that
   (a) every touched bundle is matched;
   (b) subject to (a), the number of identity edges in M is maximum;
   (c) subject to (a) and (b), $|M|$ is maximum.
2. If M is perfect, return $Y_i = M(i)$ for all i.
3. Otherwise, let i^* be an unmatched agent. Among all non-empty bundles, choose Z_{j^*} maximising
$$\max_{g \in Z} \; v_{i^*}(Z \setminus \{g\}) \;=\; v_{i^*}(Z) - \min_{g \in Z} v_{i^*}(g),$$
and remove from Z_{j^*} a good g^* of minimum value for i^* (the good is donated).

In words: an unmatched agent points at the bundle she would most like to have *after removing its least valuable good*, and this good is removed. The existence of a matching as in step 1 is shown in Lemma 1. Each iteration takes polynomial time, and there are at most $m+1$ iterations (Exercise 3).

Analysis

Lemma 1. *In every iteration, G has a matching that covers all touched bundles.*

Proof. Initially no bundle is touched. Suppose that in some iteration M is the matching of step 1, i^* is unmatched, and Z_{j^*} loses the good g^*; let $Z'_{j^*} = Z_{j^*} \setminus \{g^*\}$ and let G' be the new feasibility graph. We construct a matching of G' covering $T \cup \{Z'_{j^*}\}$.

(1) *(i^*, Z_{i^*}) is not an edge of G.* Otherwise, if Z_{i^*} is unmatched, add this edge to M; if Z_{i^*} is matched to some $k \neq i^*$, replace (k, Z_{i^*}) by (i^*, Z_{i^*}). Both contradict rule (b).

(2) *$j^* \neq i^*$ and (i^*, Z'_{j^*}) is an edge of G'.* By (1), Z_{i^*} is not EFX-feasible for i^*: there are k and $g \in Z_k$ with $v_{i^*}(Z_k \setminus \{g\}) > v_{i^*}(Z_{i^*})$. By the choice of Z_{j^*},

$$v_{i^*}(Z'_{j^*}) \;=\; \max_k \max_{g \in Z_k} v_{i^*}(Z_k \setminus \{g\}) \;>\; v_{i^*}(Z_{i^*}).$$

Hence $j^* \neq i^*$ (removing a good from Z_{i^*} cannot increase its value). Moreover Z'_{j^*} is EFX-feasible for i^* in the new bundles: towards Z_k with $k \neq j^*$ by the displayed maximum, and towards Z'_{j^*} itself trivially.

(3) *All other edges survive.* If (i, Z_j) is an edge of G with $j \neq j^*$, it is an edge of G': the bundle Z_j is unchanged, one other bundle became smaller (so EFX-feasibility is only easier), and if $i = j^*$ the condition $v_i(Z_j) > v_i(Z_i)$ also only got easier.

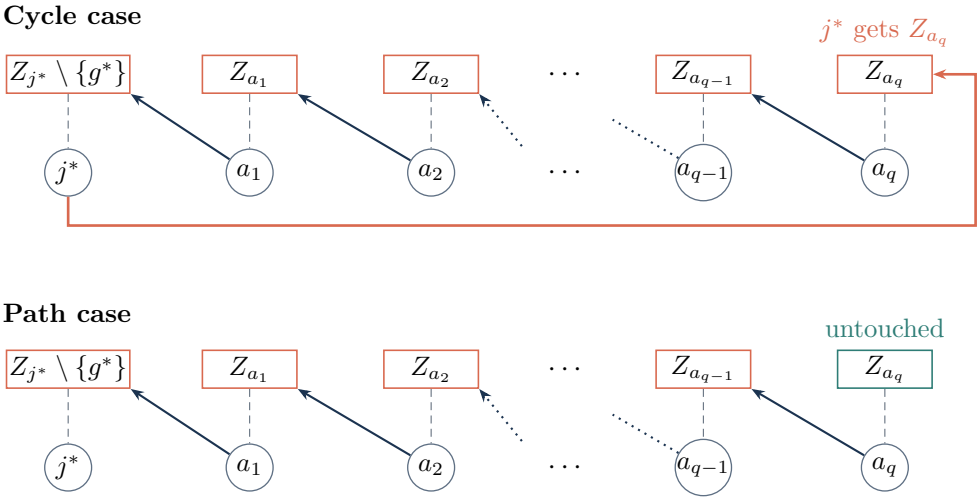
(4) *Z_{j^*} is matched in M*, say to agent k. Otherwise $M \cup \{(i^*, Z_{j^*})\}$ is a matching of G (the edge exists in G by the argument in (2)) that satisfies (a), has the same number of identity edges and is larger, contradicting (c).

Now $M' = \big(M \setminus \{(k, Z_{j^*})\}\big) \cup \{(i^*, Z'_{j^*})\}$ is a matching of G' by (2) and (3), and it covers $T \cup \{Z'_{j^*}\}$. □

Lemma 2. *Throughout the algorithm, $v_i(Z_i) \geq \frac{1}{2} v_i(X_i)$ for every agent i.*

Proof. Suppose not, and consider the first violation: in some iteration, the good g^* is removed from Z_{j^*} and $v_{j^*}(Z_{j^*} \setminus \{g^*\}) < \frac{1}{2} v_{j^*}(X_{j^*})$. Let M' be the matching of G' from the proof of Lemma 1; it covers all touched bundles, including $Z_{j^*} \setminus \{g^*\}$, and $M'(Z_{j^*} \setminus \{g^*\}) = i^* \neq j^*$. In this proof, the *current* bundle of j^* is $Z_{j^*} \setminus \{g^*\}$, and that of every other agent a is Z_a.

Follow the bundles backwards: let $a_1 = M'(Z_{j^*} \setminus \{g^*\})$ and, as long as the bundle Z_{a_r} is touched, let $a_{r+1} = M'(Z_{a_r})$; this agent exists since M' covers all touched bundles. We stop as soon as $M'(Z_{a_r}) = j^*$ or Z_{a_r} is untouched. Since M' is a matching, the agents $a_1, a_2, \ldots$ are distinct and different from j^*, so the walk ends after at most $n - 1$ steps, at some agent a_q. Every edge on the walk is a non-identity edge of G', so every agent on the walk strictly prefers the bundle she is matched to over her own current bundle. There are two cases (see the pictures).



Bundles on top, agents below. A dashed line joins an agent to her current bundle; an arrow from an agent to a bundle means that M' matches them.

Cycle: Z_{a_q} is touched and $M'(Z_{a_q}) = j^*$. Then $j^*, a_1, \ldots, a_q$ form a cycle: a_1 is matched to $Z_{j^*} \setminus \{g^*\}$, each a_{r+1} to Z_{a_r}, and j^* to Z_{a_q}. Let every agent on the cycle give up her current bundle and take her matched bundle instead:

$$\widehat{X}_{a_r} = (X_{a_r} \setminus Z_{a_r}) \cup M'(a_r) \quad (1 \le r \le q), \qquad \widehat{X}_{j^*} = \big(X_{j^*} \setminus (Z_{j^*} \setminus \{g^*\})\big) \cup Z_{a_q},$$

and let all other agents keep their bundles in X. Every agent on the cycle strictly gains and all others keep their value, so $\mathrm{NW}(\widehat{X}) > \mathrm{NW}(X)$, contradicting the maximality of X.

Path: Z_{a_q} is untouched, i.e. $Z_{a_q} = X_{a_q}$. Reallocate along the path: a_1 takes $Z_{j^*} \setminus \{g^*\}$, each a_{r+1} takes Z_{a_r}, and a_q keeps her whole bundle X_{a_q} in addition:

$$\widehat{X}_{a_q} = X_{a_q} \cup M'(a_q), \qquad \widehat{X}_{a_r} = (X_{a_r} \setminus Z_{a_r}) \cup M'(a_r) \quad (1 \le r < q), \qquad \widehat{X}_{j^*} = X_{j^*} \setminus (Z_{j^*} \setminus \{g^*\}),$$

and all other agents keep X. Then:

- $v_{a_q}(\widehat{X}_{a_q}) = v_{a_q}(X_{a_q}) + v_{a_q}(M'(a_q)) > 2\, v_{a_q}(X_{a_q})$, since $v_{a_q}(M'(a_q)) > v_{a_q}(Z_{a_q}) = v_{a_q}(X_{a_q})$;
- $v_{a_r}(\widehat{X}_{a_r}) > v_{a_r}(X_{a_r})$ for $1 \le r < q$;
- $v_{j^*}(\widehat{X}_{j^*}) = v_{j^*}(X_{j^*}) - v_{j^*}(Z_{j^*} \setminus \{g^*\}) > \frac{1}{2} v_{j^*}(X_{j^*})$, by the violation.

Hence the product of the values increases by a factor larger than $2 \cdot \frac{1}{2} = 1$, again a contradiction. □

Lemma 3. *When the algorithm stops, at least one bundle is untouched.*

Proof. Suppose all n bundles are touched at the end. Consider the iteration in which the last untouched bundle Z_{j^*} was touched. At that time, $n-1$ bundles were touched and covered by M (Lemma 1), and Z_{j^*} was matched as well (step (4) of the proof of Lemma 1). So M was perfect, and the algorithm would have stopped instead. □

Proof of Theorem 2. By Lemma 1 the algorithm is well defined, and it stops after at most $m+1$ iterations with a perfect matching M of G; hence Y is EFX.

M is the identity matching. Otherwise M contains a cycle of non-identity edges (agents $a_1 \to a_2 \to \cdots \to a_1$, where a_r is matched to $Z_{a_{r+1}}$), and exchanging the corresponding goods as in the cycle case of Lemma 2 would increase $\mathrm{NW}(X)$. So $Y_i = Z_i \subseteq X_i$ for all i.

Nash welfare. By Lemma 2, $v_i(Y_i) \ge \frac{1}{2} v_i(X_i)$ for all i, and by Lemma 3 $Y_{i_0} = X_{i_0}$ for some agent i_0. Hence

$$\mathrm{NW}(Y)^n = \prod_i v_i(Y_i) \;\ge\; 2^{-(n-1)} \prod_i v_i(X_i) \;=\; 2^{-(n-1)}\, \mathrm{opt}^n. \qquad \square$$

Remarks.

- Computing an MNW allocation is NP-hard (Exercise 2), so the theorem is not an efficient algorithm by itself. Caragiannis, Gravin and Huang show that their algorithm can be modified to start from a ρ-approximate MNW allocation instead; it then computes, in polynomial time, an EFX partial allocation whose Nash welfare is within a factor 2ρ of opt. Combined with a constant-factor approximation algorithm for Nash welfare, this gives a polynomial-time constant-factor approximation by an EFX partial allocation.
- In the theorem, the donated goods may be many, and agents may envy the set of donated goods. Chaudhury, Kavitha, Mehlhorn and Sgouritsa (2020) later showed that one can always donate fewer than n goods such that the allocation is EFX and nobody envies the donated goods (for monotone valuations).

Exercises

Question 1 (The factor is tight). Show that the factor $2^{-(1-1/n)}$ in Theorem 2 cannot be improved: for every n and $\delta > 0$ there is an instance with additive valuations in which every EFX partial allocation Y satisfies $\mathrm{NW}(Y) < \left(2^{-(1-1/n)} + \delta\right) \mathrm{opt}$.

Question 2 (Computing MNW is hard).

(a) Show that deciding whether an instance has an allocation with $\mathrm{NW} \geq k$ is NP-hard, even for $n = 2$ agents with the *same* additive valuation.

(b) Show that it is strongly NP-hard for n agents with the same additive valuation.

Question 3 (Running time). Show that, given X, the algorithm of Caragiannis, Gravin and Huang runs in polynomial time:

(a) the feasibility graph G can be computed in polynomial time;

(b) a matching as in step 1, with priorities (a), (b), (c), can be computed in polynomial time;

(c) the algorithm stops after at most $m + 1$ iterations.

Hints.
Q2: (a) Reduce from Partition: given positive integers $a_1, \ldots, a_m$, is there $T \subseteq [m]$ with $\sum_{j \in T} a_j = \sum_{j \notin T} a_j$?
(b) Reduce from 3-Partition (strongly NP-hard): given positive integers $a_1, \ldots, a_{3n}$ with $\sum_j a_j = nB$ and $B/4 < a_j < B/2$, can they be split into n groups, each of sum B?